Certified Tail Bounds for Prime-Power Threshold Cutoff Sequences

Abstract

For an integer level $k \geq 1$, put

$$\text{powerTailSum}(x, k) = x + x^2 + \cdots + x^{k+1}.$$

If $t \geq 83$ is prime and u is the next prime after t, define the prime cutoff $C_k(t)$ to be the largest prime q, or 0 if none exists, such that

$$\text{powerTailSum}(q, k) < \text{powerTailSum}(u, 4).$$

Thus $C_4(t) = t$. This paper isolates the threshold-83 tail-budget theorem

$$\sum_{k \geq m} C_k(t) \leq t - \min(3m + 1, 2m + 21, 83) \qquad (m \geq 13).$$

The proof has an elementary analytic handoff and a finite exact-prefix certificate. The analytic side is written out in full. The finite side is packaged as a frozen certificate and an independent deterministic verifier. The verifier recomputes all finite-prefix rows by exact integer arithmetic; its acceptance is the computational proof of the finite part.

1 Introduction

This paper studies a concrete family of prime cutoff sequences. Given a prime t, let u be the next prime. The level-k cutoff $C_k(t)$ is the largest prime whose level-k prime-power tail lies below the level-4 threshold determined by u. The main point is that, once $t \geq 83$, the late cutoffs have an explicit threshold-row budget:

$$\sum_{k \geq m} C_k(t) \leq t - \min(3m + 1, 2m + 21, 83) \qquad (m \geq 13).$$

The statement is elementary in form. It involves only primes, finite geometric sums, and an explicit finite threshold certificate. Its proof does not use the Riemann hypothesis, Robin's criterion, or Lagarias's criterion. Those criteria explain why this cutoff sequence arose, but they are not needed for the theorem as stated. For that reason the paper first develops the cutoff sequence on its own terms and only later records the divisor-sum motivation.

The proof splits into two parts. The first part gives a uniform analytic bound for all sufficiently large threshold rows. The second part verifies the remaining finite prefix exactly. The large-row handoff is compact enough to write as integer inequalities; the finite prefix is much larger, so it is handled by an exhaustive verifier using a frozen threshold certificate. The accompanying package records the nineteen handoff rows in `certificate.csv`. The verifier recomputes the finite prefix from those rows without using exploratory search code.

The contribution is therefore deliberately narrow. First, the paper isolates the cutoff sequence as an independent elementary object. Second, it proves a uniform analytic handoff reducing all large rows to nineteen integer inequalities. Third, it supplies a finite-prefix verifier for the remaining rows. This is the shape expected of a computational number theory result: the mathematics reduces the infinite statement to a finite exhaustive check controlled by a small threshold certificate, and the certificate can be checked without trusting the search that found it.

2 Definitions and main theorem

Definition 1 (Power-tail sum)**.** *For an integer $k \geq 1$ and a real number x, define*

$$\mathrm{powerTailSum}(x, k) = \sum_{j=1}^{k+1} x^j.$$

When $x = q$ is prime, $\mathrm{powerTailSum}(q, k)$ *is the integer prime-power tail used to define the cutoff at level k.*

Definition 2 (Threshold-83 cutoff sequence)**.** *Let $t \geq 83$ be prime, and let u be the least prime greater than t. For $k \geq 1$, define*

$$C_k(t) = \max\bigl(\{0\} \cup \{q \in \mathcal{P} : \mathrm{powerTailSum}(q, k) < \mathrm{powerTailSum}(u, 4)\}\bigr).$$

We usually write C_k when t is fixed. Since $\mathrm{powerTailSum}(x, 4)$ *is strictly increasing and there is no prime strictly between t and u, we have $C_4(t) = t$.*

Theorem 1 (Threshold-83 exact tail-budget staircase)**.** *Let $t \geq 83$ be prime, let u be the next prime after t, and define $C_k = C_k(t)$ as above. Then for every integer $m \geq 13$,*

$$\sum_{k \geq m} C_k \leq t - \min(3m + 1, 2m + 21, 83).$$

Remark 1. *The theorem is certificate-backed by the accompanying package. The proof below prints the analytic handoff and specifies the threshold certificate precisely. The finite prefix is not printed row by row and is not claimed to be a single self-contained Lean theorem; instead,* `verify_tail_certificate.cpp` *recomputes the finite prefix exhaustively from the frozen certificate by exact integer arithmetic.*

3 Proof strategy

The proof begins with a one-line consequence of the cutoff definition. If $C_k > 0$, then

$$\mathrm{powerTailSum}(C_k, k) < \mathrm{powerTailSum}(u, 4).$$

The lower bound $\mathrm{powerTailSum}(q, k) > q^{k+1}$, the upper bound $\mathrm{powerTailSum}(u, 4) \leq 2u^5$, and Bertrand's postulate $u < 2t$ imply

$$C_k^{k+1} < 64t^5.$$

Therefore

$$C_k < (64t^5)^{1/(k+1)}.$$

There are also only $O(\log t)$ nonzero levels, because $C_k > 0$ forces $2^{k+1} < 64t^5$. Summing the uniform envelope over $k \geq m$ gives

$$\sum_{k \geq m} C_k \leq 64^{1/(m+1)}(6 + 5\log_2 t)t^{5/(m+1)}.$$

For $t \geq 256$, the elementary logarithmic envelope

$$6 + 5 \log_2 t \leq 36t^{1/4}$$

turns this into a pure power inequality. For each $13 \leq m \leq 31$, it then suffices to verify one integer handoff row:

$$36^{4a} 64^4 T^{20+a} \leq (T - g_m)^{4a}, \qquad a = m + 1, \quad g_m = \min(3m + 1, 2m + 21, 83).$$

The exponent in the corresponding ratio is less than 1, so once the inequality holds at the handoff T, it holds for every larger t. The finite range $83 \leq t < T$ is checked exactly from the cutoff definition. For $m \geq 31$, monotonicity of the tail in m reduces the result to the case $m = 31$, where the staircase has already reached the constant gap 83.

4 Proof of the theorem

Lemma 1 (Active cutoff envelope)**.** *Let $t \geq 83$ be prime and u the next prime. If $C_k(t) > 0$, then*

$$C_k(t)^{k+1} < 64t^5.$$

Proof. By definition,

$$\mathrm{powerTailSum}(C_k, k) < \mathrm{powerTailSum}(u, 4).$$

For $q \geq 2$, $\mathrm{powerTailSum}(q, k) > q^{k+1}$. Also

$$\mathrm{powerTailSum}(u, 4) = u + u^2 + u^3 + u^4 + u^5 \leq 2u^5$$

for $u \geq 2$. Bertrand's postulate gives $u < 2t$. Hence

$$C_k^{k+1} < 2u^5 < 2(2t)^5 = 64t^5.$$

□

Lemma 2 (Smooth tail majorant)**.** *For every $m \geq 1$,*

$$\sum_{k \geq m} C_k(t) \leq 64^{1/(m+1)} (6 + 5 \log_2 t) t^{5/(m+1)}.$$

Proof. If $C_k > 0$, then $C_k \geq 2$, so Lemma 1 gives

$$2^{k+1} < 64t^5.$$

Thus

$$k + 1 < 6 + 5 \log_2 t.$$

For every nonzero term with $k \geq m$, Lemma 1 also gives

$$C_k < (64t^5)^{1/(k+1)} \leq (64t^5)^{1/(m+1)}.$$

There are fewer than $6 + 5 \log_2 t$ possible nonzero terms. Multiplying the number of possible terms by the largest displayed bound gives the result. □

Lemma 3 (Quarter-root analytic handoff)**.** *Let $13 \le m \le 31$, put $a = m+1$, and put*

$$g_m = \min(3m+1, 2m+21, 83).$$

Assume $T \ge 256$ and

$$36^{4a} 64^4 T^{20+a} \le (T - g_m)^{4a}. \tag{1}$$

Then for every $t \ge T$,

$$\sum_{k \ge m} C_k(t) \le t - g_m.$$

Proof. For $t \ge 256$, the elementary estimate

$$6 + 5 \log_2 t \le 36 t^{1/4}$$

holds. Indeed, at $t = 256$ the right side is 144, while the left side is 46, and the difference is increasing for $t \ge 256$ because

$$\frac{d}{dt}\left(36t^{1/4} - 6 - 5\log_2 t\right) = \frac{1}{t}\left(9t^{1/4} - \frac{5}{\log 2}\right) > 0.$$

Combining this estimate with Lemma 2 gives

$$\sum_{k \ge m} C_k(t) \le 36\, 64^{1/a} t^{5/a + 1/4}.$$

The right side is at most $t - g_m$ exactly when

$$36^{4a} 64^4 t^{20+a} \le (t - g_m)^{4a}. \tag{2}$$

The ratio of the left side of (2) to the right side is a constant times

$$\frac{t^{20+a}}{(t - g_m)^{4a}}.$$

Equivalently, after taking $4a$-th roots, it is enough to show that

$$36\, 64^{1/a} t^{(20+a)/(4a)} \le t - g_m.$$

Put $r = (20+a)/(4a)$. For $m \ge 13$ we have $a \ge 14$, hence $0 < r < 1$. The function

$$\frac{t^r}{t - g_m}$$

is strictly decreasing for $t > g_m$, since

$$\frac{d}{dt} \log \frac{t^r}{t - g_m} = \frac{r}{t} - \frac{1}{t - g_m} = \frac{(r-1)t - r g_m}{t(t - g_m)} < 0.$$

Thus once (1) holds at T, (2) holds for all $t \ge T$. The repository formalizes this integer monotonicity statement as the quarter-root handoff. The claimed tail bound follows. ∎

Proposition 1 (Finite-prefix certificate)**.** *For $13 \leq m \leq 31$, define $g_m = \min(3m+1, 2m+21, 83)$. The following handoff thresholds satisfy the hypotheses of Lemma 3:*

m	13	14	15	16	17	18	19	20	21	22
T_m	19596	10674	6641	4542	3332	2578	2079	1733	1481	1293

m	23	24	25	26	27	28	29	30	31
T_m	1149	1037	947	873	813	763	721	685	654

Moreover, for every prime t with $83 \leq t < T_m$, the exact cutoff sequence satisfies

$$\sum_{k \geq m} C_k(t) \leq t - g_m.$$

Proof. The first assertion is a finite list of integer inequalities of the form

$$36^{4a} 64^4 T_m^{20+a} \leq (T_m - g_m)^{4a}.$$

These inequalities are small enough to be checked directly by integer arithmetic and are checked by the packaged verifier. The same handoff rows are also mirrored by the repository's Lean handoff table.

The second assertion is the finite exact-prefix certificate. The verifier recomputes the successor prime u, recomputes each $C_k(t)$ from the inequality

$$\mathrm{powerTailSum}(C_k, k) < \mathrm{powerTailSum}(u, 4),$$

sums the tail, and checks the displayed inequality. No floating-point arithmetic is needed for this finite prefix. In the present paper this finite row list is not printed; Section 5 specifies the packaged certificate and verifier that audit it independently. □

Proof of Theorem 1. First suppose $13 \leq m \leq 31$, and put

$$g_m = \min(3m + 1, 2m + 21, 83).$$

If $83 \leq t < T_m$, Proposition 1 gives

$$\sum_{k \geq m} C_k(t) \leq t - g_m.$$

If $t \geq T_m$, Lemma 3 gives the same inequality from the integer handoff row. Hence the theorem holds for $13 \leq m \leq 31$.

If $m \geq 31$, then

$$\sum_{k \geq m} C_k(t) \leq \sum_{k \geq 31} C_k(t) \leq t - 83.$$

Since

$$\min(3m + 1, 2m + 21, 83) = 83 \qquad (m \geq 31),$$

this is exactly the asserted bound. □

5 Certificate and verifier architecture

The theorem is well suited to referee-grade computational verification because the threshold certificate can be made small in concept and deterministic in execution. The verifier should not depend on exploratory search code. It should accept only a frozen certificate and either accept or reject it by direct arithmetic.

The computation package consists of four audit files:

certificate	`certificate.csv`
verifier	`verify_tail_certificate.cpp`
instructions	`BUILD_AND_VERIFY.txt`
hashes	`SHA256SUMS`

The finite part is used in the following acceptance form: the verifier `verify_tail_certificate`, compiled with the stated compiler version and invoked by the stated command line, accepts the threshold-certificate file with the stated SHA256 digest; acceptance implies Proposition 1. This is stronger than citing an exploratory repository script, because the proof obligation becomes the verifier's acceptance theorem.

The version-1.0.0 verification artifact is archived at Zenodo. Its versioned DOI is recorded in the repository metadata. The artifact uses split licensing; its file-level license map is in `LICENSE.md`.

The threshold certificate should contain the following data.

1. The nineteen handoff rows (m, T_m, g_m) for $13 \le m \le 31$.
2. A file hash for the certificate, a hash for the verifier source, and the exact command line used to reproduce the acceptance result.

The verifier should perform the following checks.

1. Recompute primality of every t and recompute that u is the least prime greater than t.
2. Recompute powerTailSum$(u, 4)$ and each cutoff $C_k(t)$ using integer arithmetic. For a positive cutoff C_k, check
$$\text{powerTailSum}(C_k, k) < \text{powerTailSum}(u, 4),$$
and either $C_k = 0$ or the next prime after C_k fails the same inequality.
3. Recompute the tail sum for every prime t with $83 \le t < T_m$ and check
$$\sum_{k \ge m} C_k(t) \le t - g_m.$$
4. Check the handoff inequalities
$$36^{4a} 64^4 T_m^{20+a} \le (T_m - g_m)^{4a}, \qquad a = m + 1.$$
5. Check the formula $g_m = \min(3m + 1, 2m + 21, 83)$ for every listed m.

The packaged verifier is deliberately independent of the exploratory scripts that found the thresholds. It uses its own prime sieve and unsigned base-10^9 big-integer arithmetic, and it uses no floating-point arithmetic.

The archived artifact keeps the finite audit path deliberately small: the standalone C++ verifier is the shipped executable specification of the finite prefix. No exploratory threshold-generation script, floating-point estimate, or external proof-assistant project is needed to check the certificate in the archive.

The minimal verifier is only a few logical steps. In pseudocode, for each listed pair (m, t) it should do the following:

$$
\begin{aligned}
&\text{assert } t \text{ is prime and } 83 \le t < T_m,\\
&u \leftarrow \text{least prime greater than } t,\\
&P_4 \leftarrow \text{powerTailSum}(u, 4),\\
&\text{for } k = m, m+1, \ldots \text{ while } \text{powerTailSum}(2, k) < P_4 :\\
&\qquad C_k \leftarrow \max\{0\} \cup \{q \in \mathcal{P} : \text{powerTailSum}(q, k) < P_4\},\\
&\text{assert } \textstyle\sum_{k \ge m} C_k \le t - g_m.
\end{aligned}
$$

The loop is finite because $C_k > 0$ implies $\text{powerTailSum}(2, k) < P_4$. This pseudocode is intentionally independent of any heuristic row generation. It is the verification contract a referee would need to trust.

6 Motivation from divisor-sum extremal analysis

The cutoff sequence above arose from divisor-sum extremal computations, but the theorem itself is independent of that origin. In Robin- and Lagarias-type analyses one often studies integers

$$n = \prod_{p \le P} p^{a_p}$$

through the normalized divisor factors

$$A_p(a) = 1 + \frac{1}{p} + \cdots + \frac{1}{p^a}.$$

A one-step marginal comparison has the form

$$\frac{A_p(a+1)}{A_p(a)} = 1 + \frac{p-1}{p(p^{a+1}-1)}.$$

After clearing denominators, this naturally produces sums of prime powers. For example, the integer quantity

$$p + p^2 + \cdots + p^{a+1}$$

is the reciprocal scale of the marginal gain. Cutoffs defined by inequalities between such power tails are therefore a natural bookkeeping device for organizing which prime powers remain active in an extremal envelope.

Classically, Robin's criterion and Lagarias's criterion connect divisor-sum inequalities with the Riemann hypothesis, and the superabundant and colossally abundant frameworks of Alaoglu and Erdos explain why exponent windows and marginal comparisons appear. The present theorem should not be read as an RH result. Its role is narrower: it gives an explicit certified tail budget for one family of prime-power threshold cutoffs that arose inside that broader extremal-number analysis.

7 Significance and limitations

The useful feature of Theorem 1 is its exactness. The right side is not merely asymptotic; it is a concrete staircase with sharp threshold-row behavior in the available data. The theorem also isolates a clean computational object: a finite prefix plus a transparent analytic handoff.

The main limitation is formalization status. The analytic handoff is elementary and the finite prefix is certified by the packaged verifier, but the submitted artifact is still a computational proof package rather than one small end-to-end proof-assistant theorem named exactly as Theorem 1.

A second limitation is scope. The theorem concerns one threshold, namely 83, and one power-tail cutoff model. It does not by itself prove a Robin, Lagarias, or Riemann-hypothesis statement. Its value is as a reusable explicit tail bound inside extremal divisor-sum analysis and as a standalone example of a certified prime-power cutoff staircase.

References

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